Daniel Engelberg

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Herzliya, Israel | Authorized to work in the US | Relocating to New York

SUMMARY

Applied AI engineer (B.Sc., Technion) who builds agents for work where mistakes cost money: a permission-aware expense agent in TypeScript with an auditable run log, durable approval chains and decision evals (zero false approvals across 3 Claude models), plus a production WhatsApp AI platform in Python with 350+ tests.

SKILLS

AI: Claude API (tool use, structured output, vision), agent loops, evals, prompt-injection defenses, OpenAI, OpenRouter, Claude Code (daily)
Programming: TypeScript, Python, JavaScript, Node.js, SQL
Full stack: Next.js, React, PostgreSQL (Drizzle, Alembic), Flask, REST APIs, webhooks, job queues, Vitest, pytest
Cloud: Vercel, Neon, Render, Google Cloud Run, Docker, Sentry, GitHub auto-deploy
Languages: English (native), Hebrew (fluent)

EDUCATION

Technion – Israel Institute of Technology, Haifa, Israel Mar 2026
B.Sc. in Mechanical Engineering, Robotics specialization
Relevant coursework: Python, Robotics (Kinematics, Path Planning), Control Theory, Signal Processing, CAD, Mechanical Design, Manufacturing Processes, Fluid Mechanics, Solid Mechanics

PROJECTS

Expense Agent (permission-aware AI expense review) | *TypeScript, Next.js, PostgreSQL, Claude API, Zod* Oct 2026

- Built an agent that reviews expense receipts: Claude vision extracts each receipt into a Zod-validated schema, a policy engine in code decides (limits, tip cap, alcohol, currency, duplicates, approval chains), and an LLM reviewer can only add human approval, never approve or change amounts
- Scoped every agent tool to the permissions of the person it acts for, so a lookup of a colleague's salary is denied and logged; nobody approves their own expense; every model call, tool call and approval attempt lands in an auditable run log with tokens, cost and latency
- Built decision evals over 42 synthetic receipts, graded on outcome and approvers with false approvals counted first: Claude Opus 5.5 and Sonnet 5.5 42/42, Haiku 5.5 40/42, zero false approvals, 3/3 prompt injections held; an eval-driven prompt fix took Sonnet from 37 to 42
- Ran each claim as a durable state machine on a Postgres job queue (SKIP LOCKED, retries with backoff, crash recovery from the last committed stage) with pause-and-resume approval chains across manager, IT and Finance
- Deployed on Vercel and Neon with demo guardrails (daily and per-visitor claim caps, nightly reset) and 108 automated tests; live at expense-agent-seven.vercel.app

Handy AI | *Python, Flask, PostgreSQL, Claude / LLM APIs, WhatsApp Cloud API* Oct 2026 – Present

- Built a WhatsApp AI maintenance coordinator for property managers: tenants report issues by text, photo or voice note, and the system runs intake, collects contractor quotes, schedules visits and requests payment
- Integrated Claude (Anthropic API) and OpenRouter with Pydantic-validated structured output, versioned prompts and per-call cost tracking; model output never changes state directly
- Engineered a Postgres-backed job queue (retries, dead-letter, per-conversation ordering, deduplication) and a 13-state ticket lifecycle with a full audit trail
- Wrote 350+ automated tests across 18,000+ lines of Python; deployed the web service and background worker on Render

EXPERIENCE

FlowTransit, Tel Aviv, Israel Jul 2026 – Sep 2026
Intern

- Improved GTFS and OpenStreetMap preprocessing so complete intracity bus lines are kept whenever any segment crosses a selected area, using map matching and network edge processing
- Studied 4 route-planning approaches (reinforcement learning with PPO, genetic algorithms, simulated annealing, shortest-path methods) to compare how each evaluates transit networks

Formula Student Technion, Haifa, Israel Oct 2023 – Apr 2024
High Voltage Engineer

- Co-designed a new high-voltage accumulator for the race car from scratch in SolidWorks, managing files in Kenesto and tasks in monday.com